

Sean Yépez · ME344 Summer 2026 · Final Project, Option 2

github.com/seanyeppez/me344ricechem

Problem: grading consumes the time meant for teaching

I love teaching, but for every 8 hours I spend teaching, 7 go to grading. I want to invert that ratio. Treemarks is my AI grading system for STEM classes: every decision checked against a rubric tied to the underlying logic, physics, and math.

AI checks every rubric decision → the TA reviews uncertain cases → students receive feedback sooner

RICECHEM · THE SUPERVISED BENCHMARK

RiceChem is the benchmark recent papers use: a chemistry exam graded against instructor rubrics. 1,264 responses · 4 questions · 27 rubric items → 8,392 TA-labeled TRUE/FALSE decisions. Agreement is measured against the TA's labeling.

- [Sonkar et al. \(2024\)](#) · introduces RiceChem and rubric-based grading of long-form answers
- [Rao & Callison-Burch \(2026\)](#) · Autorubric, rubric-based LLM evaluation
- [Ferrer et al. \(2026\)](#) · when to trust LLM graders

RESEARCH QUESTIONS

RQ1 · Quality: can tuning make a local model accurate enough for rubric grading?

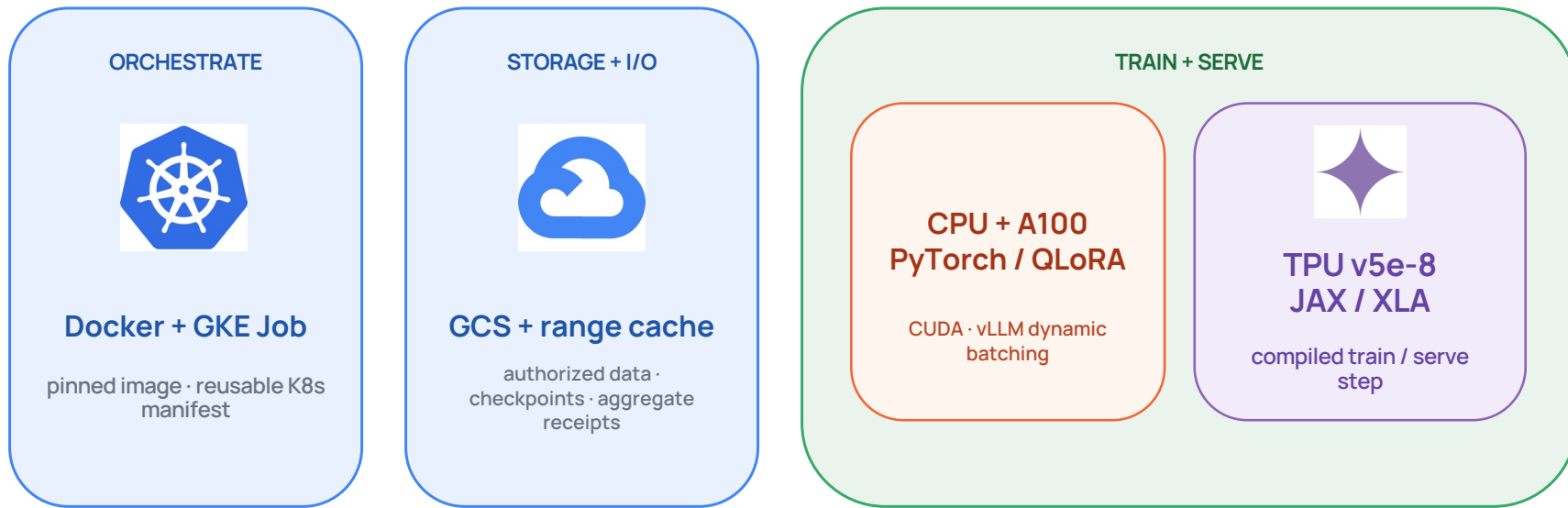
RQ2 · Systems: how do CPU, GPU, and TPU change latency, throughput, and utilization?

RQ3 · Operations: which configuration meets the quality bar in under 24 hours at lowest cost?

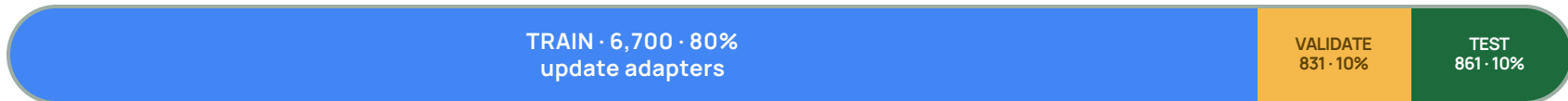


Sean Yépez · Dr. Mourad Bouache ·
Google

Containerize and run a supervised experiment on three compute lanes



EXPERIMENT DESIGN · 8,392 TA-LABELED DECISIONS



Validation picks the checkpoint. The test set is opened once, at the end.

Measurements across three kinds of compute

CONTROLLED WORKLOAD

861 held-out rubric decisions

same manifest · prompt · parser · output cap · checkpoint

concurrency 1 + 24

EXECUTION LANES

RUNTIME

CAPTURED SIGNALS

CPU

16 vCPU baseline

Transformers

host memory · process telemetry

CPU utilization

peak RSS

p50 / p95

decisions / s

GPU

one A100 40 GB

PyTorch · CUDA · vLLM

NVML · batching · endpoint timing

GPU utilization

peak VRAM

load time

tokens + decisions / s

TPU

v5e-8 slice

JAX / XLA · vLLM-TPU

XProf · compile cache · GCS-FUSE

TPU utilization

peak HBM

compile time

cache + I/O

WORKLOAD

861 rubric decisions from the
RiceChem test split · c=1 and c=24

COMPUTE

wall time · p50/p95 · decisions/s ·
wall/step time as cycle proxy

MEMORY

CPU mCPU + RSS · A100 NVML + VRAM ·
TPU host memory only

INPUT / I/O

GCS range cache fixed startup; load
and XLA compile time not isolated

Accelerators cut single-stream grading time 30–41×

THROUGHPUT SPEEDUP · CONCURRENCY 1 · CPU = 1×

16-vCPU



0.28/s · 50.9 min · 95.1% CPU · 18.0 GB RSS

A100 40 GB



8.54/s · 100.8 s · 19.5% GPU · 35.6 GB VRAM

TPU v5e-8



11.39/s · 75.6 s · device util/HBM unavailable · 54 GB host

CONCURRENT SERVING · c = 24

A100 123.9/s

\$0.007 per 861 decisions

TPU 98.4/s

\$0.023 per 861 decisions

GEMMA 4B QUALITY

The 4B agrees only ≈72% of the time,
below the grading bar.

Speed alone does not make it
deployable.

CONCLUSION

What ME344 changed for AI grading at Stanford Civil Engineering

From local prototype to cloud-hosted. At the start of the course, grading ran on a student laptop. Now inference runs in a hosted Google Cloud solution, with controlled ML experiments testing subject-specific models.

Throughput closes the feedback loop. Without local-LLM limits and with room to scale, grades that typically take a TA a week or more could come back as fast as the next morning.

Quality gates model usage. Smaller models do not offer the accuracy grading needs: at roughly 72% agreement, TAs would re-review too much. The fine-tuned 27B reached 83%, competitive with frontier models at about 1/9 the serving cost.

The operating rule. Among graders whose accuracy and throughput meet the standard, choose the cheapest.

NEXT Serve the 27B on TPU · test transfer to unseen questions · estimate the economics of training custom models versus frontier inference.